

7 (a) α : helium nucleus

β : electron

γ : electromagnetic radiation / wave / ray or photon

three correct 2 / 2, two correct 1 / 2

B2 [2]

(b) (i) atomic number / proton number / $Z - 2$, nucleon / mass number / $A - 4$

B1 [1]

(ii) atomic number / proton number / $Z + 1$
nucleon / mass number / A no change

B1 [1]

(iii) no change in proton or mass number
or "no change"

B1 [1]